

Sharaj Kunjar (they/she)

PERSONAL INFORMATION	PhD Student 232H, Network Science Institute 177 Huntington Ave, Northeastern University Boston, MA - 02115	E-mail: kunjar.s@northeastern.edu sharaj.kunjar@gmail.com Google Scholar Personal Website
RESEARCH INTERESTS	Computational Social Science, Mass communication, Language modeling, Qualitative text analysis, Complex systems, Queer theory.	
AFFILIATIONS	Network Science Institute (NetSI), Experiential AI Institute (EAI) Communication Media and Marginalization lab (CoMM)	
EDUCATION	PhD in Network Science (Social Science track) Northeastern University Advisors: Dr. Samuel V Scarpino and Prof. Brooke Foucault Welles	September 2023 - present (GPA: 4.0/4.0)
	Bachelor of Science (Research) with major in Physics In first class with distinction Indian Institute of Science, Bangalore	August 2018 - April 2022 (CGPA: 8.6/10)
SKILLS	• Programming: R, Python, MATLAB, C, $\text{\LaTeX}$, Git, HTML/CSS. • AI/ML: LLM APIs (Ollama, Anthropic, Huggingface CLI), prompt engineering, fine-tuning and agentic workflows, NLP libraries (transformers, nltk, spaCy). • Qualitative analysis: NVivo, QualCoder, content analysis, codebook development. • Big data analysis: network analysis (igraph, networkx, gephi), visualization (ggplot2, matplotlib), webscraping, bayesian inference, hypothesis testing.	
AWARDS AND HONORS	Recipient of the Kishore Vaigyanik Protsahan Yojana (KVPY) fellowship Department of Science and Technology, Government of India.	2018 - 2022
	Matter to Life (MtL-URO) undergraduate research fellowship, Max Planck school.	2021
	Gold Medal, iGEM Competition Member of IISc's team SynShine, at the International Genetically Engineered Machines jamboree.	2019
PUBLICATIONS	Articles Kunjar, S.* , Strandburg-Peshkin, A., Giese, H., Minasandra, P., Sarkar, S., Jolly, M. K., & Gradwohl, N. (2023). Link updating strategies influence consensus decisions as a function of the direction of communication. Royal Society Open Science, 10(6), 230215 https://doi.org/10.1098/rsos.230215 . Kunjar S.* , Smith A., Mckenzie T., Mohbe R., Scarpino S., Foucault Welles B. (2025). Computational frame analysis revisited: On LLMs for studying news coverage. https://arxiv.org/abs/2511.17746 Book chapters	

Foucault Welles, B., Kumar, S., **Kunjar, S.** [In Press]. Together We Can Move Mountains: Understanding the Power of Social Media Networks to Shape Democratic Deliberations. The Routledge Companion to Digital Media and Democracy.

PROFESSIONAL EXPERIENCE

Graduate Research Assistant

Network Science Institute, Boston

Developed mixed-methods pipelines integrating LLM-assisted analysis with traditional qualitative coding to study media discourse on public health crises. Evaluated GPT, Llama and Claude models against human annotators across 4000+ news articles.

September 2023
- present

AI consultant

Boston Housing Authority, Boston

Designed user-centered web resource for digital literacy curriculum through stakeholder interviews, iterative prototyping, and feedback incorporation.

January
- April 2025

Technical Assistant

Max Planck Institute of Animal Behavior, Konstanz

Processed large-scale acoustic and spatial datasets to analyze collective decision-making in meerkat groups. Contributed to Royal Society Open Science publication.

July 2022
- July 2023

Project assistant

Collective Behavior Cluster (CASCb), Konstanz

Implemented mean-field simulations modeling consensus formation on online social networks with heterogeneous user preferences and network topologies.

July
-December 2021

Research intern

Cancer Systems Biology lab, IISc Bangalore

January
- April 2022

Research intern

Soft Matter lab, IISc Bangalore

August
- December 2020

CONFERENCE PRESENTATIONS

McKenzie, T., **Kunjar, S.**, Smith, A., Trujillo, M., Foucault Welles, B., Lange, R. (2025, December 11-12). Analyzing journalistic reporting reveals blind spots in the coverage of anti-data center activism [Panel Talk], Computation+Journalism Symposium, Miami, USA.

Kunjar, S., Scarpino, S., Foucault Welles, B. (2025, June 12-16). Intermedia frame-building networks during the Mpox epidemic [Panel Talk], Mass Communication division group, 75th International Communication Association conference (ICA 2025), Denver, USA.

Kunjar, S., Scarpino, S., Foucault Welles, B. (2025, June 12-16). Intermedia frame-building networks during the Mpox epidemic [Panel Talk], International School and Conference on Network Science (NetSciX 2025), Denver, USA.

Kunjar, S., Scarpino, S., Foucault Welles, B. (2024, July 17-20). Characterizing and measuring politicization in the news media coverage of Mpox [Panel Talk], 10th International Conference on Computational Social Science (IC2S2 2024), Philadelphia, USA.

Kunjar, S., Strandburg-Peshkin, A., Giese, H., Minasandra, P., Sarkar, S., Jolly, M. K., & Gradwohl, N. (2023, April 25-28). Link updating strategies influence consensus decisions as a function of the direction of communication [Panel Talk], 14th International Conference on Complex Networks (CompleNet 2023), Aveiro, Portugal.

RESEARCH
IN PROGRESS

Framing queer health: news media and the Mpox epidemic

Investigating the politicized nature of the national news media coverage of the 2022 Mpox epidemic in the United States using frame analysis.

Can LLMs perform interpretive content analysis?

Interrogating the effectiveness of large language models (LLMs) in performing content analysis with interpretive epistemic goals using ideals of scientific objectivity.

Competitive framing drives counter-partisan intermedia influence patterns between news outlets

A network analysis of influence patterns in media framing between news outlets reveals that when the framing environment is competitive, influence is overwhelmingly cross-partisan.

COURSES

Applied mathematics, physics and engineering

Network Science I and II, Dynamical processes on complex networks, Machine Learning on graphs, Probability and Statistics, Real analysis I and II, Information theory, Dynamical systems theory, Statistical physics I and II, Classical mechanics I and II, Quantum mechanics I and II, Algorithms and programming, Neural signal Processing, Bayesian inference

Social Sciences

Social Networks, Communication and Inclusion, Queery theory, AI in media industries, Philosophy of Science, Survey methods

Biology and Behavioral Sciences

Stochastic and spatial dynamics in biology, Introduction to animal behaviour, Quantitative ecology: Research design and inference, Theoretical and mathematical ecology, Dynamical systems biology.

EXTRA
CURRICULARS

Outreach Manager

Managed social media handles and wrote content for IISc's iGEM (international genetically engineered machines) team of 2019, SynShine. Our team won a gold medal for automating optics-regulated optimal growth of bacterial co-cultures.

Artist Manager

Led a team to collaborate with Airbus, PayTM and BookMyShow to organize concerts and comedy shows at Pravega 2019.

Vocalist and Guitarist

Alumni of IISc's Music club - Rhythmica